Research Proposal Template

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Title: Simulation-to-Hardware Validation of Transformer-PPO for Disturbance Resistant Landings

Abstract

<Summarize the key points of research proposal in 200-400 words. Include highlights from introduction, background, research questions, methodology, and conclusions section of the proposal.>

This research aims to investigate the simulation-to-hardware of a gated transformer-Xl (GTrXL) enhanced Proximal-Policy-Optimization (PPO) algorithm for thrust-vectoring-control (TVC) in disturbance resistant retro-propulsive landings. Motivated by the increasing need for rocket booster recovery in reusable launch vehicles like SpaceX's Falcon 9. Current systems relies on traditional Proportional-Integral-Derivative Controller (PIDs), which struggled with dynamic disturbances like wind, sensor noise and varying initial conditions. Reinforcement-learning (RL) approaches like PPO offer promise for handling uncertainty, however standard implementations suffer with short term memory. The proposed GTrXL enhanced PPO aims to leverage gated attention mechanism for improved temporal context, enabling long term decision making. Validation of GTrXl was previously conducted in simulation, this research aims to bridge the gap between simulation-to-hardware transfer of GTrXL by running the algorithm on an electric ducted fan (EDF) drone test bed.

Background reveals literature gaps: foundational studies like Schulman et al. (2017) on PPO and Dai et al. (2019) on Transformer-XL prove simulation efficacy over PID, and Hwangbo et al. (2017) applies PPO to drones. However, transformer-augmented PPO for hardware in thrust-vectoring-control (TVC) landing systems lacks hardware validation, with the EDF drone test bed emulating TVC landing.

The three primary research questions are as follows: (1) What is the fidelity of simulation to hardware transfer for a GTrXL-PPO on an TVC EDF Drone test bed, measured by metrics like landing dispersion, jerk rate, touchdown velocity and success rate. (2) How robust is GTrXL to external disturbances in the transonic landing regeime (e.g., wind, shifting center of mass, fuel slosh, sensor noise/error and initial position), measured by metrics like recovery time, trajectory efficiency (Simulated Fuel consumed). (3) How does GTrXL-PPO compare to baselines like PPO only and PID only, with comparison metrics like successful landing rate, initial condition spread, touchdown accuracy, fuel margins and jerk.

The research adopts a pragmatic worldview and design science approach benefiting reusable vehicle recovery and crew lander safety, with a focus on producing an artifact, drone Test Bed. This research will rely on quantitative simulations training and initial evaluations, followed by hardware in the loop simulation and finally follow by flight tests in a controlled environment. Data collection involves collection of training outcomes per epoch, followed by state vector and sensor logs in the hardware in the loop and flight test phase. Statistical comparison between controllers

will be conducted using statistical methods (E.g., T-tests). Ethics is a primary focus, ensuring no harm via tethered tests, safety envelopes and disclaimers on dual-use risks (E.g., military applications under ITAR regulations)

In conclusion, this research bridges the gap between simulation and reality, demonstrating that GTrXL PPO's potential for safer, cost-effective landings. Expected contributions include verifiable RL policies for small scale proxies, paying the way for scaled up demonstrators.

Keywords

- Proximal Policy Optimization (PPO)
- Gated Transformer XL (GTrXL)
- Reinforcement Learning (RL)
- Retro-Propulsive Landing
- Thrust-Vectoring Control (TVC)
- Proportional Integral Derivative Controller (PID)
- Simulation to Hardware Validation
- Electric Ducted Fan (EDF)

Introduction/Background

< A brief overview of the proposed research topic and the motivation for its selection.>

The proposal aims to explore the simulation-to-hardware transfer of a Gated Transformer-XL (GTrXL) enhanced Proximal Policy Optimization (PPO) algorithm which enabled disturbance resistant retro-propulsive landings in reusable launch vehicles. The primary motivation of the topic are recent rapid developments in multi-reuse launch systems, driven by mega constellations like the Starlink and Amazon's LEO, NASA's Commercial off the Shelf (COTS) Artemis lunar Program and recent proposals for in-space data centers. The key accelerator for these projects is the lowering cost of access to space in commercial space through rapid reusability. In recent years, reusable launchers have been developed and put into operation by companies like SpaceX (Falcon 9 & Starship) , Blue origin's New Glenn. Various other smaller entities like Rocket Lab and Chinese firms like land Space are developing their own reusable systems.

A rockets first stage booster accounts for approximately 60% of the manufacturing cost of a launch system, which fuel comprising only about ~<1% of the total launch cost (Bushnell & Moses, 2018). The ability to reuse boosters via retro-propulsive landing enables significant cost reductions, with potential cost reductions by a factor of 14 or greater (Bushnell & Moses, 2018). It's analogous of flying on an airplane than discarding the vehicle after a single trip. To Improve operational cadence, we must expend its cross range capability, involving handling of wider range of conditions and disturbance for success recovery.

Traditional Control systems such as Proportional-Integral-Derivative (PID) controllers face fundamental limitations when applied to dynamic non-linear systems (Konstantopoulos et al., 2020). These limitation include sensitivity to parameter variations, hardware variations and disturbances like noise, wind gusts and varying initial conditions. Modern guidance and control systems (GNC) for reusable rocket employ advanced optimization-based approaches, such as sequential convex programming (SCP) to handle non-linear dynamics. However, this method too faces challenge of linearizing and discretizing the problem. To ensure convergence, SCP algorithms restrict the solution to a small "trust region" around the previous trajectory. If the vehicle needs to make a highly nonlinear maneuver (e.g., a rapid attitude slew) due to a large variation of initial condition, the required change might lie outside this trust region. The solver then reports "Infeasible" not because the rocket can't do it, but because the algorithm fails to find a solution to converge, leading to a rapid unscheduled disassembly (RUD) of the booster (Açıkmeşe & Ploen, 2007).

Reinforcement leaning (RL) methods, Particularly PPO, have demonstrated potential in managing non-linear dynamics and unforeseen trajectories due to varying initial conditions or disturbances, yet they are constrained by limited short-term memory, preventing effective long-term decision-making in complex long temporal trajectories like entry, descent and landing (EDL). By integrating gated attention mechanisms, this study aims to improve PPO’s temporal context, facilitating in more robust control in TVC landing scenarios. Recent work has demonstrated GTrXL-PPO's effectiveness: Federici et al. (2024) achieved 98.7% landing success under uncertainties in a simulated lunar landing, while Carradori et al. (2025) adapted it for simulated atmospheric rocket landings. Additional simulation validation shows capability for multi-phase trajectory optimization (Chen et al., 2024)

However as mentioned, these prior validations have been confined only to simulations, which represents a critical gap in hardware integration and flight tests. The research hence aims to bridge the current gap by integrating the GTrXL-PPO algorithm in an electric Ducted Fan (EDF) drone test bed, which use control fins in the thrust segment to emulate rocket thrust vector control, thereby bridging simulation-to-reality transfer. This topic is driven by its alignment with current aerospace advancements and future missions like NASA’s Artemis program where precise autonomous landings in challenging lunar polar environments are paramount for astronaut safety and mission success.

A. Benefits of Study

<State the contribution expected from your research efforts, what individuals/organizations will use the results of your thesis/APP and what problems/issues you feel will be addressed/resolved.>

The research on GTrXl is expected to yield several key benefits and contributions to the commercial space sector. Primarily, the research aims to bridge the critical simulation to real hardware gap in reinforcement learning (RL) for aerospace control systems, providing empirical evidence of GTrXL-PPO efficacy through quantifiable metrics such as landing success rates, touchdown accuracy, jerk rates and robustness to disturbances. The experimental results in verifiable RL policies deployable on a small scale TVC proxy like the EDF drone testbed, serving as a foundational artifact in the research, aligned to design science. This artifact shall pave the way for scaled-up demonstrators, such as the SpaceX's "Falcon Hopper" style systems. Additionally, the study will offer comparative insights into GTrXL-PPO vs baselines like standard RL PPO and PID controllers, highlighting improvements (if any) in trajectory efficiency, fuel majoring and overall system reliability.

The results of these research will be valuable to a range of organizations. Aerospace engineers and researches in institutions like NASA's Langley research center or universities focused on RL can use the validated policies and methodologies to advance autonomous guidance systems. Commercial space entities such as SpaceX , Blue Origin rocket lab and emerging US players can benefit form the insights to enhance reusable launch vehicle recovery, reducing operational cost, increasing reliability and crew safety. These commercial entities also support NASA particularly in their COTS program, which provides the human landing system for NASA's project Artemis. These findings can be used as a stepping stone to improve precision landings in challenging polar lunar environments, thereby enhancing crew safety. Broader stakeholders that include constellation operators (e.g., Starlink and amazon LEO), may leverage improved reliability and launch cadence to drive operational cost lower.

For Research Community: This work establishes a replicable methodology for hardware validation of transformer-based RL policies using low-cost test platforms (EDF drones). The quantitative comparison against state-of-the-art SCP methods and traditional PID controllers fills a critical gap in the literature, providing empirical benchmarks for future attention-based algorithm development

In conclusion, this benefits several pressing problems, including a quantitative benchmark against traditional PID controllers in handling nonlinear dynamics and external perturbation. It also resolves the literature gap in hardware validation of transformer-augmented RL for thrust-vectoring control (TVC), mitigating risks associated with simulation-only approaches, such as poor transfer fidelity. Ultimately,

it tackles issues of cost inefficiency in expendable rockets (e.g., boosters representing ~60% of manufacturing costs) and safety concerns in human-rated landers, fostering more robust, autonomous, and economical space transportation systems.

B. Research Questions

<Identify the primary research subsidiary research questions. The primary research question/topic should be broad enough that it covers the entire spectrum of the research activity. Subsidiary research questions subdivide the primary research question into manageable research segments. This should be a very explicit and concise statement of the questions the research will seek to answer. While the questions may be redefined later as the research progresses, the initial objectives should be made specific.>

Question 1: What is the fidelity of simulation-to-hardware transfer for a GTrXL–PPO controller on an EDF drone, as measured by landing dispersion, jerk rate, touchdown velocity, and landing success rate under controlled conditions?

This addresses the gap in hardware demonstrations and simulation transfer, with measurable metrics for quantification.

Plausible target metrics for a EDF testbed for n landing attempts:

Metric	Definition	Plausible Target (EDF test Bed)	Statistical Validation
Landing Dispersion Epsilon (ε)	Circular Error Probable - radius of circle containing 50% of landing points (meters)	<0.1m	Power Analysis: n=100 (Landings) p=0.8 alpha = 0.05
Jerk	Second order Acceleration (m/s^3)	<10m/s^3	ANOVA
Touchdown Velocity	Vertical and horizontal velocity (m/s)	<0.5m/s	ANOVA
Landing Success Rate	Percentage of success landing	>99%	N = 100 (landings)
Controller Latency	Sensor input to control output (ms)	<50ms	General max latency for real time control.
Trajectory Efficiency (Delta V)	Magnitude of velocity changes	n/a	Lower the better

Question 2: How robust is the GTrXL–PPO controller to disturbances such as varying initial conditions, wind, sensor noise, and shifting center of mass, evaluated by performance metrics in both simulation and EDF drone hardware tests?

This targets the gap in evaluating transformer-based PPO policies on realistic hardware, with specific, testable disturbance factors and measurable outcomes.

Plausible target metrics for a EDF testbed for n landing attempts:

Metric	Definition	Plausible Target (EDF test Bed)	Statistical Validation
Trajectory Deviation	Root-mean-square RSME deviation from nominal path (meters)	<0.1m	Monte Carlo (n=100)
Recovery Time	Duration to return within 5% of nominal trajectory after disturbance (seconds)	n/a	Lower the better n=100
Control Effort	Maximum thrust magnitude and simulated gimbal rate (N, rad/s)	Below Saturation Limits	
Robustness Margin	Maximum disturbance magnitude before failure	Monte Carlo	Sensitivity analysis

Question 3: How does the performance of the GTrXL–PPO controller compare to baseline controllers (e.g., standard PPO, PID and SCP) in terms of touchdown accuracy and safety-margin metrics (e.g., fuel remaining) during simulation and EDF drone landings?

This focuses on the gap in NN verification to GTrXL–PPO on flight hardware (Small scale proxy), using specific comparative metrics for evaluation.

Plausible target metrics for a EDF testbed for n landing attempts: Using Similar metric to Question1:

Hypotheses

Hypotheses 1: The GTrXL-PPO controller will demonstrate high-fidelity simulation-to-hardware transfer on the EDF drone testbed, achieving a landing success rate of >99% and touchdown dispersion of <0.1m under nominal conditions (Addressing Research Q1).

Hypotheses 2: The GTrXL-PPO will exhibit superior robustness to disturbances (e.g., wind gusts up to 10 m/s and sensor noise) compared to baselines. (Addressing Research Q2).

Hypotheses 3: The GTrXL-PPO will outperform baseline controllers (standard PPO, PID, and SCP) in key metrics, including a improvement in successful landing rate and fuel margins during both simulation and hardware tests (Addressing Research Q3).

C. <u>Scope of the Project</u>

<Describe the main thrust of the study, what areas will be specifically investigated and what areas will be excluded. Put boundaries around the study. Identify what the study will be (e.g., a case study, implementation guide, development of a model,

assessment of a model, policy or management guide). Discuss any limitations of the study.>

The scope of this study focuses on simulation-to-hardware transfer for existing GTrXL-PPO algorithm with a focus on emulating transonic retro-propulsive landing phase in a EDF drone testbed.

The study includes the following scope:

1. Deployment of existing GTrXL, PPO, PID and SCP algorithms for 6-DOF landing control (Source code available).
    a. Training in Nvidia's Issac Sim levering its high-fidelity physics simulation environment
    b. Development of custom value and policy functions (GTrXL-PPO and PPO only)
    c. Hyperparameter tuning of all algorithms (Automatic during training, Ziegler-Nichols method for PID)
2. Setup of Simulation environment
    a. 6 DOF rigid body model, with dynamic center of mass simulation for TVC.
    b. Disturbance Models: Wind fields, sensor noise, mass & center of mass variations.
    c. Simulated landing terrain (landing pad)
3. Hardware Platform
    a. FMS 90mm Metal Ducted Fan with 12-Blade (~45N of thrust), Thrust to weight >1.5
    b. 4 servo fins to simulate Thrust vector control
    c. 2x Inertial measurement unit (BNO085), WTGAHRS1 10-axis High-stability IMU AHRS, PX4 optical flow camera
    d. Embedded Nvidia Jetson Nano
    e. All commercial off the shelve hobby grade hardware
4. Experimental Validation
    a. Hardware in the loop testing (Via MATLAB Simulink integration with Nvidia Issac Sim)
    b. Tethered flight tests. 100+ controlled landing tests (To meet research question metrics)
    c. Disturbance Injection: Simulated wind using a fan and variable payloads including fuel slosh (Container of Water as payload on EDF)
5. Baselines Validated:
    a. Sequential Convex Programming (SCP)
    b. Standard PPO w/o long short-term memory (LSTM)
    c. Ziegler-Nichols tuned PID controller
    d. Statistical comparison using t-tests/ANOVA (alpha = 0.05)
6. Not included in Study:
    a. Full scale hardware test including actual rocket boosters.
    b. No untethered flight tests.

    c. Does not include reentry descent phase, including atmospheric entry (hypersonic, subsonic and transonic regime), analogous final 1-2km of rocket landing.
    d. Research prototype only, not intended for deployment on crewed or commercial systems
    e. Not intended for use for military purposes.
7. Limitations and Boundaries:
    a. Scaling Challenges:
        i. EDF drone dynamics differ from full-scale rockets (e.g., higher thrust-to-weight ratio: 3:1 vs. rocket's ~1.2-1.5:1)
        ii. Negligible fuel mass variation (electric vs. propellant consumption)
        iii. Different Reynolds number and aerodynamic scaling
        iv. Different moment of inertia.
    b. Environmental Constraints
        i. Indoor testing limits maximum wind disturbance to ~10 m/s (simulated via fans)
        ii. Temperature and pressure remain at sea-level nominal
    c. Sensor Fidelity
        i. Consumer-grade IMU (<0.16°/min heading error with calibration Dynamic Error: 3.5°, Static Error: 2.0°, BNO085 IMU)
        ii. Consumer-grade GPS/Barometer, improved with optical flow sensor (1Hz, 0.1m accuracy)
        iii. Noise characteristics may differ from operational spacecraft systems
        iv. No radiation effects or space environment simulation
    d. Computational Platform
        i. Embedded computing (Jetson Nano) representative of small spacecraft but not large boosters
        ii. Small Sim-to-real gap expected due to quantization and platform differences
    e. Validation Scope
        i. Results demonstrate feasibility for small unmanned vehicles only
        ii. Extrapolation to larger vehicles requires additional validation
8. Expected Outcomes:
    a. Partial Open source GTrXL-PPO implementation with trained weights (Hugging Face)
    b. Isaac Sim environment source code for reproducible training on Github
    c. Hardware test bed design (Bill of materials) documentation and CAD files
    d. Dataset of >100 hardware flight tests with state vectors

NASA Technology Readiness Level (TRL): This research advances GTrXL-PPO from TRL 3 (analytical/experimental proof-of-concept) to TRL 5 (technology validated in relevant environment), establishing readiness for integration into larger demonstration platforms.

Literature Review

2. Literature Review

2.1 Foundations of Traditional Optimization-Based Guidance

Early work on powered descent guidance laid the groundwork for handling nonlinear dynamics in planetary landings, emphasizing convex optimization to ensure real-time feasibility. Açıkmeşe and Ploen (2007) introduced a convex programming approach for Mars landing, framing the powered descent problem as a sequence of convex subproblems that linearize nonlinear constraints within trust regions. This sequential convex programming (SCP) method achieves convergence by restricting solutions to feasible neighborhoods around prior trajectories, enabling precise control under fuel constraints and initial dispersions. However, as noted by the authors, large deviations such as rapid attitude changes due to disturbance can exceed trust regions, leading to infeasibility and potential mission failure. For aerospace GNC, this highlights SCP's strengths in deterministic environments but limitations in highly nonlinear or uncertain scenarios, motivating RL alternatives that do not rely on explicit linearization.

Steinfeldt et al. (2010) examined GNC performance trades for pinpoint Mars landing, evaluating sensor suites, navigation filters, and control algorithms against metrics like touchdown accuracy and fuel margins. Their analysis revealed that traditional methods, including proportional-integral-derivative (PID) controllers and optimization-based guidance, struggle with sensor noise, wind disturbances, and varying initial conditions, often requiring trade-offs in precision versus computational load. In retro-propulsive contexts, such as Falcon 9-style boosters, these trades underscore the need for robust feedback laws that integrate long-term state histories, paving the way for RL policies that learn adaptive behaviors from simulation data.

2.2 Foundations of Proximal Policy Optimization in Reinforcement Learning

Building on traditional methods, RL offers a data-driven paradigm for optimizing policies in complex, uncertain environments. Schulman et al. (2017) proposed Proximal Policy Optimization (PPO), a policy gradient algorithm that improves upon trust region policy optimization by using clipped surrogate objectives to constrain policy updates, ensuring stable learning in high-dimensional continuous control spaces. PPO's simplicity and sample efficiency make it suitable for tasks with sparse rewards or partial observability, achieving state-of-the-art performance on benchmarks like robotic locomotion. In aerospace applications, PPO's ability to handle nonlinear dynamics without explicit model assumptions addresses SCP limitations, such as infeasibility under large disturbances, by iteratively refining policies through simulated interactions.

Hwangbo et al. (2017) demonstrated PPO's practical utility in hardware, applying it to quadrotor control for agile maneuvers. Their approach trained policies in simulation

and transferred them to real drones, mitigating sim-to-real gaps via domain randomization of disturbances like sensor noise and wind. Results showed PPO outperforming model predictive control in recovery time and trajectory efficiency, though hardware constraints highlighted challenges in real-time inference. For this research, Hwangbo et al.'s work motivates using PPO as a baseline for electric ducted fan (EDF) testbeds, where transformer enhancements could further improve long-horizon reasoning for retro-propulsive landing under varying conditions.

2.3 Transformer-XL

Dai et al. (2019) proposed Transformer-XL to overcome the fixed-length context limitation of the vanilla Transformer in language and context modeling. Their main innovation is a segment-level recurrence mechanism that reuses hidden states from previous segments as a form of extended memory, allowing the model to capture dependencies over thousands of time steps without recomputing all attention scores. They also introduced a relative positional encoding scheme that decouples position information from absolute sequence length, improving both generalization and efficiency. Empirically, Transformer-XL achieves state-of-the-art perplexity on several long-context language modeling benchmarks and trains faster than comparable baselines (Dai et al., 2019).

For RL and control applications, especially in aerospace, the key insight from Transformer-XL is that explicit long-term memory can stabilize learning and improve performance when decisions depend on events far in the past, such as earlier vehicle states, thrust commands, or attitude disturbances. Although Dai et al. (2019) focused on language tasks, the architectural ideas segment recurrence and relative positional encoding are directly applicable to partially observable Markov decision processes (POMDPs) that arise in flight control. This motivates subsequent work that adapts Transformer-XL to RL settings and investigates how attention-based memory can support robust control policies.

For aerospace GNC, Transformer-XL could enhance PPO-based controllers by incorporating a long context window over sensor readings and estimated vehicle states. Such memory may help the controller reason about slow-varying disturbances like fuel usage or unintended environmental perturbations, thereby improving landing accuracy and robustness against real-world uncertainties like aerodynamic noise, engine plume interactions, or sensor dropouts. At the same time, vehicle-side hardware is often limited; the quadratic cost of self-attention and the added complexity of recurrent memory highlight the need for architectural refinements and careful verification before deployment in safety-critical flight software.

2.4 Survey on Transformers in Reinforcement Learning

Li et al. (2023) surveyed the rapidly growing field of transformers in reinforcement learning, organizing existing work into several categories, including sequence modeling for decision-making, trajectory-based offline RL, model-based planning, and generalist agents that operate across multiple tasks. They catalogued how attention mechanisms are used to encode histories of states, actions, and rewards, and how transformer architectures are adapted to handle continuous control, sparse rewards, and multi-task learning.

A key contribution of the survey is its analysis of when transformer-based RL methods provide clear benefits over more traditional architectures such as recurrent neural networks (RNNs) or convolutional networks. Li et al. (2023) highlighted that transformers excel when long-range temporal credit assignment and flexible context aggregation are important, for example, in tasks with continuous or episodic structure, while also noting limitations in computational cost, data requirements, and real-time reliability.

For this research, the survey clarifies where PPO-style transformer policies are most likely to add value: in long-duration, partially observable control problems such as powered descent landing (PDL), where the agent must integrate information over many time steps and adapt to changing or unknown conditions. At the same time, Li et al. (2023) pointed out that safety, interpretability, and formal certification remain underexplored in transformer-based RL. This reinforces the need to combine architectural advances like Gated Transformer-XL (GTrXL) with systematic verification and validation methods before considering deployment on actual aerospace vehicles.

2.5 Stabilizing Transformers for Reinforcement Learning (GTrXL)

Parisotto et al. (2020) proposed Gated Transformer-XL (GTrXL) to address training instabilities that arise when applying Transformer-XL directly to RL tasks. They introduced gating mechanisms and architectural refinements that help control gradient flow and mitigate exploding or vanishing gradients, issues that plague LSTM and RNNs. In a range of benchmark environments, GTrXL consistently outperformed both LSTM-based agents and vanilla Transformer-XL policies, especially on tasks that require long-term memory and robust credit assignment.

The authors demonstrated that GTrXL not only improves asymptotic performance but also stabilizes learning dynamics, reducing the number of catastrophic policy collapses and training restarts. From the perspective of aerospace control, this stability is crucial in retro-propulsive landing, where margins are thin, and the policy must reliably converge in simulation before it can be considered for any form of hardware testing. GTrXL therefore serves as a more practical architecture for safety-critical RL applications than the original Transformer-XL.

In the broader context of this review, GTrXL forms the critical bridge between general-purpose sequence models and domain-specific applications such as powered landing. It shows that transformer-style memory can be made trainable and robust in challenging RL settings, but it does not, by itself, solve questions of safety, verification, or certification. Those issues become central for aerospace applications and motivate the need to connect GTrXL-based controllers with the verification and validation frameworks surveyed by Zhang and Li (2020).

2.6 Transformer-Based Robust Feedback Guidance for Atmospheric Powered Landing

Carradori et al. (2025) applied transformer-based meta-RL, specifically a gated Transformer architecture, to the problem of atmospheric powered landing with retro-propulsive engines. Their guidance law was trained in a moderate-fidelity simulation environment that captures semi-realistic aerodynamics, varying wind profiles, navigation errors, and dispersed initial conditions representative of planetary entry scenarios. The resulting policy generates thrust and vectoring commands that achieve accurate touchdowns across a wide envelope of initial states and disturbance realizations.

Methodologically, the authors combined meta-RL (e.g., PPO-based training over a distribution of scenarios) with transformer memory to encode trajectories, allowing the controller to adapt online to different initial conditions and disturbances (Carradori et al., 2025). All heavy computation is performed offline during training, while online inference reduces to a sequence of forward passes through the trained network, making the approach compatible with real-time flight computers in principle. This design illustrates how transformer-based controllers can be engineered so that most computational cost is incurred before flight.

From a systems viewpoint, Carradori et al. (2025) demonstrated that transformer-based meta-RL can deliver robust feedback guidance under complex, uncertain atmospheric conditions, but their work remains limited to simulation and does not include hardware nor provide a full safety case for flight software. For example, there is little discussion of rare-event testing, formal verification of closed-loop behavior, or integration with existing certification standards. These gaps connect directly to the testing and verification challenges that are the focus of Zhang and Li’s (2020) systematic review and highlight where additional work is required before similar controllers could be deployed on an EDF drone testbed.

2.7 Systematic Review of Testing and Verification for NN-Based Safety-Critical Control

Early efforts in verifying neural network (NN)-based controllers for aerospace emphasized runtime assurance and formal methods to mitigate risks in safety-critical systems. Luckall et al. (2002) provided a foundational NASA framework for

verification and validation (V&V) of NNs, focusing on reachability analysis, adversarial testing, and integration with legacy flight software. Their guidelines stressed the importance of bounding NN behaviors under uncertainties like sensor faults, offering practical checklists for aerospace deployment while noting challenges in scaling formal proofs to complex architectures.

Building on this, Zhang and Li (2020) conducted a systematic literature review of testing and verification techniques for neural network–based safety-critical control software. They catalogued a wide spectrum of methods, including formal verification (e.g., reachability analysis and SMT-based approaches), testing frameworks for adversarial inputs and rare events, monitoring and runtime assurance architectures, and interpretability tools. The review emphasized gaps in real-time applicability and hardware integration, particularly for RL policies where nondeterminism complicates certification.

For this research, these works underscore the need to apply V&V to GTrXL-PPO controllers, extending simulation successes to EDF hardware through metrics like robustness to disturbances and comparative baselines. By addressing these, the proposed EDF testbed can bridge foundational V&V principles with modern transformer-RL, advancing toward larger scale testbeds.

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Methodology

<This section should explain the conceptual foundation or framework within which the questions will be examined and the specific research techniques that will be used to answer the questions. A description of how the goal is to be achieved: how the hypothesis will be tested, how the experiment will be conducted, etc. Provide enough detail so that an assessment can be made of the feasibility and suitability of the proposed work. Identify the sources from which the data will be collected, and the methodology to be used in collecting the data. Be specific in identifying the method(s) of research.>

This research adopts a pragmatic worldview, which emphasizes practical applications and solutions to real-world problems. This approach is particularly suitable for addressing the challenges of reusable launch vehicle recovery and crew safety in aerospace systems, where the focus is on actionable outcomes. The methodology follows a design science research framework (Hevner et al., 2004), which prioritizes the creation and evaluation of artifacts to solve identified problems. The primary artifact is an Electric Ducted Fan (EDF) drone test bed that emulates Thrust-Vectoring Control (TVC) for retro-propulsive landings. The study also employs a primarily quantitative approach with elements of mixed methods. The research will proceed in three sequential phases: (1) simulation-based training and evaluation, (2) hardware-in-the-loop (HIL) simulation, and (3) controlled flight tests.

Phase 1: Simulation-Based Training and Initial Evaluation

The GTrXL-PPO algorithm will be trained in a high-fidelity simulation environment using NVIDIA's Isaac Sim, which supports realistic physics modeling for 6-DOF (degrees of freedom) dynamics, including basic aerodynamics, thrust vectoring, and disturbances like wind gusts (up to 10 m/s), sensor noise (Gaussian noise with $\sigma=0.1$), shifting center of mass (±10% variation, typical for a Falcon 9 scale vehicle during landing), and varying initial conditions (e.g., altitude 5-10m, velocity 0-5 m/s). The simulation will leverage open-source implementations of GTrXL (Parisotto et al., 2020) and PPO (Schulman et al., 2017), adapted for TVC control. Training will use a meta-RL approach.

Data sources include generated simulation logs: state vectors (position, velocity, orientation, thrust commands), reward signals per epoch, and performance metrics (e.g., landing dispersion in meters, jerk rate in m/s^3, touchdown velocity in m/s, success rate as binary outcome). Training episodes will be run per controller variant (GTrXL-PPO, standard PPO, PID, and Sequential Convex Programming [SCP]

baselines). Data collection will be automated via Python scripts interfacing with Issac Sim's API

Initial evaluations will compare controllers using paired t-tests to assess statistical significance ($p<0.05$) in metrics like trajectory efficiency (simulated fuel consumed, normalized to 0-1) and recovery time (seconds to stabilize after disturbance).

Phase 2: Hardware-in-the-Loop (HIL) Simulation

The trained GTrXL-PPO policy will be deployed on an embedded flight controller (NVIDIA Jetson Nano) integrated with the EDF drone test bed. The drone features a FMS Model 90mm EDF with servo-actuated control fins for TVC emulation, inertial measurement units (IMUs), GPS, and optical flow. HIL testing will use a real-time simulator (MATLAB Simulink) to feed synthetic sensor data to the hardware while monitoring outputs.

Data collection involves logging state vectors and sensor data at standard 100 Hz via onboard telemetry, capturing the same metrics as in simulation plus latency (ms). Approximately 500 HIL trials will be conducted per algorithmic variant, introducing controlled disturbances. Analysis will use ANOVA for multi-controller comparisons and R-Squared calculated for simulation-to-hardware transfer correlation.

Phase 3: Controlled Flight Tests

Final validation occurs through tethered flight tests in a controlled indoor environment (e.g., a 10x10m typical classroom on campus). The EDF drone will perform autonomous descents from 5-10m altitudes, with disturbances introduced via external fans (wind), added weights (container of water), and sensor noise injection. Flight test will be conducted on GTrXL-PPO controller only due to time and hardware limitations.

Data will be collected via onboard sensor logs and onboard optical flow camera (for ground truth via optical flow tracking, ground marker to denote landing pad), yielding ~100 flights for GTrXL-PPO controller. Metrics will be analyzed using statistical tests as in prior phases, with confidence intervals (95%) reported to quantify uncertainty. Feasibility is supported by low-cost hardware (<$500 for EDF setup) and adherence to safety protocols (e.g., Teather and software limits).

Overall, this methodology is feasible given accessible tools (open-source RL libraries like Stable Baselines3, simulation platforms) and a timeline of 6-9 months.

Evaluation of Results

Outcomes will be evaluated quantitatively against targets derived from literature (e.g., Federici et al., 2024; Carradori et al., 2025) and various simulation benchmarks. Success criteria include: (1) simulation-to-hardware transfer fidelity >90% (measured by metric correlation via Pearson's r); (2) robustness under

disturbances with success rates >99%; (3) comparative superiority over PPO/PID/SCP in metrics like touchdown accuracy (<0.5m dispersion) and fuel margins (>20% remaining, simulated). Qualitative evaluation will involve review of logs for emergent behaviors (e.g., unforeseen emergent adaptive recovery strategies)

Results will be visualized via plots (e.g., trajectory in Matplotlib) and tables that display different landing attempts and final touchdown parameters.

Ethical Considerations

Although my project primarily relies on simulated data and a small EDF (electric ducted fan) drone, there are still ethical considerations related to safety, data integrity, and responsible use of autonomous control systems. Below, I provide a comprehensive review, drawing on relevant studies and frameworks from AI ethics, aerospace engineering, reinforcement learning (RL) and dual use.

The primary ethical concern arises during the hardware experimentation phase, where the autonomous controller can have unsafe behaviors, such as erratic flight paths leading to crashes. This aligns with the principle of "no harm" in research ethics, as uncontrolled drone, powered by a 90mm EDF with fan blades rotating at over 40,000 rpm could pose risks to researchers or bystanders. Furthermore, ingestion of foreign debris can lead to a rapid unscheduled disassembly (RUD) of the fan, potentially resulting in flying debris. In aerospace contexts, where systems like retro-propulsive landings are safety-critical, failures could have amplified consequences if scaled up. To address this, drone hardware tests will be conducted in a controlled environment with the drone tethered to prevent runaway behavior and to minimize the chance of injury or property damage, I will also include manual kill switches, manually defined flight safety envelopes, and pre-test checklists.

- Studies highlight the importance of these measures. For instance, a review by the National Aeronautics and Space Administration (NASA) emphasizes verification and validation (V&V) processes for neural network-based aerospace systems to mitigate risks, noting that unverified autonomous controllers can lead to catastrophic failures in real-world deployments (Luckall et al., 2002). Similarly, a study on ethical challenges in RL for robotics stresses the need for "safety envelopes" during testing, such as physical constraints, to prevent harm, drawing from incidents like drone malfunctions in urban testing environments (Amodei et al., 2016). By incorporating these, the research ensures ethical compliance while advancing safe AI integration.
- The broader ethical dimension to deploying advanced autonomous guidance systems in aerospace is safety and reliability. For my experiment, ethical risks include potential biases in simulation parameters (e.g., unrealistic disturbances) that could lead to misleading results, or data manipulation during analysis to overstate performance improvements. To mitigate this, all simulation data will be stored securely with version control (e.g., via GitHub), backed up regularly, and reported honestly in publications. I will use open-

source tools for reproducibility, sharing code and datasets where possible, and conduct sensitivity analyses to check for biases in metrics like position error or stability. During analysis and conclusions, I will employ statistical rigor (e.g., T-tests as outlined in the workshop) and disclose any limitations or gaps.

- A systematic review on testing neural network-based safety-critical software identifies reproducibility as a key ethical issue, citing cases where non transparent data handling in RL studies led to irreproducible claims and eroded trust in aerospace applications (Zhang & Li, 2020). Furthermore, the ACM Code of Ethics (2018) mandates transparency in computational models, particularly for systems with societal/environmental impact, to prevent "black-box" model issues in RL (Gotterbarn et al., 2018). By prioritizing openness in science, this research addresses these gaps.
- The proposal involves AI-driven reinforcement learning (RL) for disturbance-resistant landings, using simulations (e.g., NVIDIA Isaac Sim) and hardware testing on an EDF drone to improve trajectory optimization under conditions like wind or sensor noise. While framed for civilian aerospace (e.g., reusable rockets like Falcon 9 or Starship), this technology inherently supports dual-use applications, specifically military applications.

RL with attention mechanisms enables long-term context driven decision-making, which could enhance autonomous guidance in military systems. For instance, the same algorithms for rocket trajectories could optimize missile paths for evasion, targeting, or precision strikes, as RL excels in dynamic, uncertain environments like evading defenses or pursuing targets.

- Under U.S. export controls, such technologies might fall under ITAR if it is deemed "defense articles" (e.g., guidance, navigation, and control systems for rockets or unmanned aerial vehicles) (Squire Patton Boggs, n.d.). If not explicitly ITAR-controlled, they could be regulated under the Export Administration Regulations (EAR), which govern dual-use items via the Commerce Control List (CCL).
- Ethically, this dual-use aspect raises concerns about unintended use. AI technologies like RL, originally from nonmilitary fields (e.g., self-driving cars, or humanoid robots), can be repurposed for weapons, lowering barriers to advanced military capabilities and potentially escalating conflicts or enabling non-state actors.
- Ensure to include disclaimers (e.g., "Not intended for military use"), and commitments to open-source only non-sensitive aspects (Model but no weights potentially).

This research adopts design science producing an artifact and a pragmatic worldview, aligning with mixed-methods ethics guidelines from the American Psychological Association (APA), emphasizing beneficence, justice, and respect (Gotterbarn et al., 2018). No conflicts of interest exist, and institutional review board (IRB) approval will be sought if hardware tests evolve to involve human subjects. By integrating these ethical issues, the project not only addresses immediate risks but contributes to ethical standards in AI-driven aerospace research.

Appendix I

Anticipated Funding Requirements

< Be realistic in identifying your needs. The chair of your thesis/APP committee and RASP may assist in identifying sources of funds.>

Bill Of Materials:

Item	Desc	Cost	Link
FMS 90mm EDF	FMSEDF011 90mm Ducted 12-blade 4068-Kv1850 Inner Runner	~100 USD	FMS Model Website
ESC	Predator ESC 120 Amp for 3-8S LIPO	73.48 USD	PorcupineRC
LIPO	Turnigy Heavy Duty 4000mAh 8S 60C LiPo Battery Pack (XT90/EC5)	62 USD	Hobby King
Jetson Nano	128-Core Maxwell GPU, 4-Core 64-bit ARM CPU, 4GB LPDDR4, 16GB eMMC	129 USD	Authorized Reseller Site
AHRS IMU	WitMotion WTGAHRS1 10-axis High-stability IMU AHRS Inclinometer, High-precision Acceleration Gyro Angle Magnet Air Pressure GPS	118 USD	Amazon
Optical Flow Camera	Compact optical flow and distance sensor module that combines a PAA3905E1 optical flow sensor, a Broadcom AFBR-S50LV85D distance sensor	125 USD	HolyBro
IMU Backup	BNO085 IMU	24.95 USD	Adafruit
4x Servos	KST DS215MG Micro Digital Metal Gear Servo	35 USD * 4	Amain Reseller Site
Frame and Structure	Frame and structural components via RC carbon fibre Rods and 3D-printed joints	<50 USD	Hobby King
MISC	Various Misc parts and software	<100 USD	
Grand Total:		~ 860 USD	

Total Estimated Budget: < 1000USD. This is conservative, assuming reuse of university resources (e.g., MATLAB Simulink for HIL via student licenses). No personnel costs are included, as this is a solo graduate thesis.

Potential Funding Sources:

University grants: Apply for RASP (Research and Scholarship Program) funding or departmental thesis support (2,000 USD for STEM projects).

External grants: NASA EPSCoR Seed Grants or AIAA student research awards (targeting aerospace RL applications, 500-1000 USD).

Anticipated IRB and Other approvals

<List IRB and other university approvals your research may require.>

Anticipated Approvals:

Institutional Review Board (IRB) Approval:

- Not required for the core project, as it does not involve human subjects, surveys, interviews, or any form of human participation or data collection from individuals. The drone tests are fully autonomous and conducted in isolation (in an enclosed class Room), with safety measures (e.g., tethering, kill switches) to prevent harm.

University Facility and Safety Approvals:

- Campus Facility Use Permit: Required for indoor flight tests in university spaces (e.g., classrooms or labs).
- Storage and charging of LIPO batteries in a fire-resistant container on campus.
- Environmental Health and Safety (EHS) Committee Approval, compliance with OSHA standards and university policies on hazardous equipment

Export Controls and Dual-Use Compliance:

- Office of Research Compliance (ORC) Review for ITAR/EAR, Recommended due to the dual-use potential of the technology. Disclaimers (e.g., "Not intended for military use") will be included in publications

Other Potential Approvals:

- Drone Operation Permit (FAA Part 107 Waiver if Outdoors), however this can be mitigated with help from MNSU Maverick Drone racing club with certified part 107 operators.

Thesis/APP Committee

Dr Mansoor Naseef (Machine Learning RL expert)

Dr. Qun (Vincent) Zhang (RCET, control systems and sensor integration expert)

Appendix II

Tentative Timetable for Completion of the Thesis/APP

Include dates for:

- completion of research
- first draft of thesis (at least six weeks prior to the defense)
- final draft (at least two weeks prior to the defense)
- defense (confirm the date with the chair of the thesis/APP committee)

Research Project/Thesis/APP Stage	**Completion Target Date**
Literature review	Jan 2026
Finalize research questions	Jan 2026
Finalize the thesis/APP committee	Jan 2026
Construct research design	Feb 2026
Construct Issac Sim Environment	Mar 2026
Construct Article	Jul 2026
Conduct experiment/gather data	Nov 2026
Analyze data	Dec 2026
Draft thesis to advisor(s)	Jan 2027
Schedule defense	Mar 2027
Final thesis submission for signatures:	Apr 2027

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